

SOC Reporting System

Installation & Setup Guide for Windows

Version 1.0.0 Classification: Internal Use Only NIST Compliant

Table of Contents

- 1. System Requirements
- 2. Automated Installation (Recommended)
- 3. Manual Installation
- 4. Database Setup
- 5. Application Configuration
- 6. First Login & Initial Setup
- 7. Network Access via SSH/Ethernet
- 8. Daily Operations
- 9. Troubleshooting
- 10. Security Considerations

1. System Requirements

Component	Requirement
Operating System	Windows 10/11 (64-bit) or Windows Server 2019+
RAM	Minimum 4 GB (8 GB recommended)
Disk Space	Minimum 2 GB free space
Network	Ethernet interface for direct connection
Browser	Chrome, Firefox, or Edge (latest version)
Software	PowerShell 5.1+ (pre-installed on Windows 10/11)

What gets installed: XAMPP (Apache 2.4 + PHP 8.2 + MariaDB 10.4) - a single portable package that includes everything the application needs to run.

2. Automated Installation (Recommended)

The installer package includes scripts that automate the entire setup process.

1 Extract the Installer

Extract the `SOC-Reporting-Installer.zip` file to any location on the Windows machine (e.g., the Desktop).

2 Run the Installer

1. Open the extracted `SOC-Reporting-Installer` folder
2. Right-click `INSTALL.bat`
3. Select **"Run as administrator"**
4. The installer will:
 - Download XAMPP (Apache + PHP + MariaDB) automatically
 - Install it to `C:\xampp`
 - Configure PHP extensions and timezone
 - Create the database user and import all 5 databases
 - Deploy the application to `C:\xampp\htdocs\SOC-Reporting`
 - Start Apache and MariaDB services
5. Wait for the process to complete (5-10 minutes depending on internet speed)

Firewall: Windows may ask to allow Apache and MariaDB through the firewall. Click **"Allow access"** for both - these are required for the application to function.

3 Verify the Installation

1. The installer will open `http://localhost/SOC-Reporting/setup.php` in your browser
2. All items should show green checkmarks
3. If everything passes, click **"Open SOC Reporting System"**

4 First Login

Field	Value
URL	<code>http://localhost/SOC-Reporting/</code>
Username	<code>admin</code>
Password	<code>Admin@SOC2024</code>

SECURITY: Change the admin password immediately after first login via Admin Panel > Users.

IMPORTANT: Delete the file `C:\xampp\htdocs\SOC-Reporting\setup.php` after successful setup to prevent unauthorized database reinitialization.

3. Manual Installation

If the automated installer fails, follow these steps to install manually.

3.1 Install XAMPP

1. Download XAMPP from <https://www.apachefriends.org/download.html>
2. Choose the **Windows** version with **PHP 8.2.x**
3. Run the installer and install to [C:\xampp](#)
4. During installation, select at minimum: **Apache, MySQL, PHP**

3.2 Configure PHP

1. Open [C:\xampp\php\php.ini](#) in a text editor (e.g., Notepad)
2. Find and uncomment (remove the `;`) these lines:

```
extension=pdo_mysql  
extension=mbstring  
extension=openssl  
extension=gd
```

3. Find the [date.timezone](#) setting and set it:

```
date.timezone = Europe/Vienna
```

4. Save the file

3.3 Deploy Application Files

1. Copy the [SOC-Reporting](#) folder from the installer package
2. Paste it into [C:\xampp\htdocs\](#)
3. The final path should be [C:\xampp\htdocs\SOC-Reporting\](#)
4. Verify that [C:\xampp\htdocs\SOC-Reporting\index.php](#) exists

3.4 Start Services

1. Open [C:\xampp\xampp-control.exe](#) (XAMPP Control Panel)
2. Click **"Start"** next to **Apache**
3. Click **"Start"** next to **MySQL** (this is actually MariaDB)
4. Both should turn green when running

Tip: To have XAMPP start automatically with Windows, click "Config" in the XAMPP Control Panel and check "Autostart of modules": Apache and MySQL.

3.5 Create Database User

1. Open a command prompt (CMD)
2. Navigate to MariaDB:

```
cd C:\xampp\mysql\bin
```

3. Connect to MariaDB:

```
mysql -u root
```

4. Run these SQL commands:

```
CREATE USER 'soc_admin'@'localhost' IDENTIFIED BY 'SOC_Local_2024!';
GRANT ALL PRIVILEGES ON soc_system.* TO 'soc_admin'@'localhost';
GRANT ALL PRIVILEGES ON soc_machine_1.* TO 'soc_admin'@'localhost';
GRANT ALL PRIVILEGES ON soc_machine_2.* TO 'soc_admin'@'localhost';
GRANT ALL PRIVILEGES ON soc_machine_3.* TO 'soc_admin'@'localhost';
GRANT ALL PRIVILEGES ON soc_machine_4.* TO 'soc_admin'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

3.6 Import Database Schemas

Option A: Use the built-in setup page:

1. Open <http://localhost/SOC-Reporting/setup.php> in your browser
2. The setup script will create all databases and tables automatically

Option B: Import manually via command line:

```
cd C:\xampp\mysql\bin
mysql -u root < C:\xampp\htdocs\SOC-Reporting\sql\system.sql
```

For each machine database (replace N with 1, 2, 3, 4):

```
mysql -u root -e "CREATE DATABASE IF NOT EXISTS soc_machine_N"
REM Then copy machine_template.sql, replace {N} with the number, and import
```

Easier option: Just visit <http://localhost/SOC-Reporting/setup.php> in your browser - it handles all database creation automatically.

4. Database Architecture

The system uses 5 separate databases for security isolation:

Database	Purpose
soc_system	Users, sessions, audit logs, machine config, form fields, instructions
soc_machine_1	Reports and data for Machine 1
soc_machine_2	Reports and data for Machine 2
soc_machine_3	Reports and data for Machine 3
soc_machine_4	Reports and data for Machine 4

Key Tables

Table	Database	Description
users	soc_system	User accounts with bcrypt password hashes
machines	soc_system	Machine names and configuration (4 slots)
form_fields	soc_system	Dynamic form field definitions per machine
reports	soc_machine_*	Event reports with timestamps and reaction times
report_data	soc_machine_*	Dynamic field values for each report
deleted_reports	soc_machine_*	Archived deleted reports (soft delete)
audit_log	soc_system	All admin and user actions logged

5. Application Configuration

5.1 Database Credentials

File: `C:\xampp\htdocs\SOC-Reporting\config\database.php`

```
define('DB_HOST', 'localhost');
define('DB_PORT', '3306');
define('DB_USER', 'soc_admin');
define('DB_PASS', 'SOC_Local_2024!');
```

Change the database password for production use. Update both the MariaDB user password and the `database.php` config file.

5.2 Application Settings

File: `C:\xampp\htdocs\SOC-Reporting\config\app.php`

Setting	Default	Description
<code>SESSION_LIFETIME</code>	3600 (1 hour)	User session timeout
<code>ADMIN_SESSION_LIFETIME</code>	1800 (30 min)	Admin panel session timeout
<code>AUTH_MAX_ATTEMPTS</code>	5	Login attempts before lockout
<code>AUTH_LOCKOUT_TIME</code>	900 (15 min)	Lockout duration
<code>REPORTS_PER_PAGE</code>	25	Pagination size
<code>MACHINE_SLOTS</code>	4	Number of machines

5.3 Timezone

The timezone is set in `config/app.php` :

```
date_default_timezone_set('Europe/Vienna');
```

Change this to match your server's timezone if different. This must match the MariaDB timezone to prevent session expiry issues.

6. First Login & Initial Setup

6.1 Login

1. Open `http://localhost/SOC-Reporting/` in your browser
2. Login with username `admin` and password `Admin@SOC2024`

6.2 Admin Panel Setup (Required)

1. Click **"Admin Panel"** in the sidebar (bottom)
2. Enter the admin credentials again (admin authentication gate)
3. Complete these initial configuration steps:

a) Change Admin Password

1. Go to Admin Panel > **Users**
2. Edit the `admin` user
3. Set a strong new password (NIST: minimum 8 characters)

b) Configure Machine Names

1. Go to Admin Panel > **Machines**
2. Rename each machine slot to match your actual machines
3. Optionally add descriptions

c) Build Report Forms

1. Go to Admin Panel > **Form Builder**
2. Select a machine
3. Add fields: text, textarea, dropdown, checkbox, number
4. Set labels in English and German
5. Drag to reorder fields
6. Save the form configuration

d) Configure Dropdown Options

1. Go to Admin Panel > **Dropdown Data**
2. Select dropdown fields and add their options
3. Provide bilingual labels (EN/DE)

e) Create User Accounts

1. Go to Admin Panel > **Users**

2. Create accounts for SOC analysts
3. Assign role: `user` (standard) or `admin`
4. No self-registration - all accounts are admin-created

7. Network Access via SSH/Ethernet

The system is designed to run on a standalone Windows machine accessed via direct Ethernet connection and SSH.

7.1 Network Configuration

1. Connect an Ethernet cable directly between the server and client
2. On the server (Windows), set a static IP:
 - Go to Network Settings > Ethernet adapter > Properties
 - Set IP: `192.168.1.1` , Subnet: `255.255.255.0`
3. On the client, set a compatible static IP:
 - IP: `192.168.1.2` , Subnet: `255.255.255.0`

7.2 Configure Apache for Remote Access

1. Open `C:\xampp\apache\conf\httpd.conf`
2. Find `Listen 80` - keep as is (listens on all interfaces)
3. In Windows Firewall, allow incoming connections on port 80

7.3 Access from Client

On the client machine, open a browser and navigate to:

```
http://192.168.1.1/SOC-Reporting/
```

7.4 SSH Access (Optional)

To enable SSH on Windows for remote management:

1. Open Windows Settings > Apps > Optional Features
2. Add "OpenSSH Server"
3. Start the service: `net start sshd`
4. From the client: `ssh user@192.168.1.1`

8. Daily Operations

8.1 Starting the Server

Double-click `START-SERVER.bat` in the installer folder, or:

1. Open `C:\xampp\xampp-control.exe`
2. Start **Apache** and **MySQL**

8.2 Stopping the Server

Double-click `STOP-SERVER.bat` in the installer folder, or:

1. Open XAMPP Control Panel
2. Stop **Apache** and **MySQL**

8.3 Application Features

Feature	Description
Dashboard	Overview of all 4 machines, report counts, avg reaction times
Report Creation	Dynamic forms with mandatory incident time, auto-calculated reaction time
Report Filtering	Filter by date range, status, event ID, username with advanced time picker
Charts Dashboard	Splunk-style analytics: 10 chart types, KPI panels, multi-machine overlays
PDF Export	Export individual reports as formatted PDF documents
Bilingual	Full English/German support, switchable via EN/DE toggle
Soft Delete	Reports are archived, not permanently deleted. Restorable by admin
Audit Logging	All actions logged: logins, report creation, deletions, admin changes

8.4 Backup

Regular database backups are essential. Use this command in CMD:

```
cd C:\xampp\mysql\bin
mysqldump -u root --databases soc_system soc_machine_1 soc_machine_2 soc_machine_3 s
```

Recommendation: Schedule daily backups using Windows Task Scheduler. Create a batch file with the command above and schedule it to run nightly.

9. Troubleshooting

9.1 Apache Won't Start

Problem	Solution
Port 80 in use	Check with <code>netstat -aon findstr :80</code> . Stop IIS or Skype if conflicting. Or change Apache port in <code>httpd.conf</code>
Missing VCRUNTIME	Install Visual C++ Redistributable from Microsoft (vc_redist.x64.exe)
Firewall blocking	Allow <code>httpd.exe</code> through Windows Firewall

9.2 Database Connection Error

Problem	Solution
MariaDB not running	Start MySQL via XAMPP Control Panel
Wrong credentials	Verify <code>config/database.php</code> matches MariaDB user
Databases don't exist	Run <code>setup.php</code> again or import SQL manually

9.3 Admin Login Issues

Problem	Solution
Session expires immediately	Ensure PHP timezone in <code>php.ini</code> matches <code>config/app.php</code>
Account locked out	Wait 15 minutes, or clear via DB: <code>UPDATE users SET failed_attempts=0, locked_until=NULL WHERE username='admin'</code>
Forgot password	Generate new hash: <code>php -r "echo password_hash('NewPass123', PASSWORD_BCRYPT, ['cost'=>12]);"</code> and update in DB

9.4 Pages Not Loading

- Ensure the URL is `http://localhost/SOC-Reporting/` (with trailing slash)
- Check that `mod_rewrite` is enabled in Apache (uncomment in `httpd.conf`)
- Check PHP error log: `C:\xampp\php\logs\php_error_log`
- Check Apache error log: `C:\xampp\apache\logs\error.log`

10. Security Considerations (NIST Compliance)

The SOC Reporting System implements the following NIST-compliant security measures:

Control	Implementation
Password Hashing	bcrypt with cost factor 12
Session Management	Cryptographic tokens, HTTP-only cookies, strict SameSite
CSRF Protection	Per-session CSRF tokens on all forms
SQL Injection	PDO prepared statements everywhere
XSS Prevention	htmlspecialchars() on all user output
Access Control	Role-based (admin/user) with separate admin auth gate
Brute Force	Account lockout after 5 failed attempts (15 min)
Audit Logging	All security-relevant actions logged with timestamps and IPs
Security Headers	X-Frame-Options, X-Content-Type-Options, CSP via .htaccess
Directory Protection	Config, includes, and SQL directories blocked via .htaccess

Production Checklist:

- Change the default admin password immediately
- Change the database password in both MariaDB and config/database.php
- Delete setup.php after initial configuration
- Set up regular database backups
- Enable HTTPS if the system will be accessible over a network
- Review and restrict Windows Firewall rules
- Keep XAMPP/PHP updated for security patches

File Structure Reference

```
C:\xampp\htdocs\SOC-Reporting\  
  index.php           Main entry point & router  
  setup.php           Database setup (DELETE after use!)  
  .htaccess           Apache security configuration  
  config/  
    app.php           Application settings  
    database.php       Database credentials  
  includes/  
    auth.php          Authentication & CSRF  
    db.php            Database helper class  
    helpers.php       Utility functions  
    session.php       Session management  
  pages/  
    admin/            Admin panel pages  
  api/                API endpoints  
  assets/  
    css/style.css     Dark SOC theme  
    js/               JavaScript files  
  sql/  
    system.sql        System database schema  
    machine_template.sql Machine database template
```

SOC Reporting System v1.0.0 | Installation Guide

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